

10. Which of the following statements is incorrect?
- 1) Boiling point of D_2O is more than H_2O
 - 2) Dihydrogen does not act as a reducing agent
 - 3) Hydrogen has three isotopes of which tritium is the less common
 - 4) Hydrogen never acts as cation in ionic salts
11. Some statements about heavy water are given below:
- (i) Heavy water is used as a moderator in nuclear reactors
 - (ii) Heavy water is more associated than ordinary water
 - (iii) Heavy water is more effective solvent than ordinary water
- 1) (i) and (ii)
 - 2) (i), (ii) and (iii)
 - 3) (ii) and (iii)
 - 4) (i) and (iii)
12. At its melting point ice is lighter than water because:
- 1) H_2O molecules are more closely packed in solid state
 - 2) ice crystals have hallow hexagonal arrangement of H_2O molecules
 - 3) On melting of ice the H_2O molecules shrinks in size
 - 4) ice forms mostly heavy water on first melting
13. Identify the **incorrect** statement regarding H_2O_2 .
- 1) It decomposes in presence of metal surfaces and alkali present in glass containers.
 - 2) Its decomposition is catalyzed by dust particles.
 - 3) It is stored in wax lined glass or plastic vessels.
 - 4) Urea, acetanilide, stannates etc. catalyse its decomposition
14. In which of the following reactions water act as reducing agent?
- I) photosynthesis
 - II) reaction with F_2
 - III) reaction with sodium
 - IV) reaction with cupric oxide
- 1) I, II only
 - 2) I, II and III
 - 3) I, II, IV
 - 4) All
15. Which of the following has highest concentration of PAN?
- 1) Smoke
 - 2) Ozone
 - 3) Photochemical smog
 - 4) Reducing smog
16. Which one of the following is not a common component of Photochemical smog?
- 1) Ozone
 - 2) Acrolein
 - 3) Peroxyacetyl nitrate
 - 4) Chlorofluorocarbons

17. Which one of the following statements is not true?
- 1) Clean water would have a BOD value of less than 5 ppm
 - 2) Fluoride deficiency in drinking water is harmful. Soluble fluoride is often used to bring its concentration upto 1 ppm.
 - 3) When the pH of rain water is higher than 6.5, it is called acid rain
 - 4) Dissolved Oxygen (DO) in cold water can reach a concentration upto 10 ppm
18. In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300K is 25 mm, the percentage of nitrogen in the compound is
- 1) 16.76
 - 2) 15.76
 - 3) 17.36
 - 4) 18.20
19. In the Kjeldahl's method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.75g of sample neutralized 10 mL of 1M H_2SO_4 . The percentage of nitrogen in the soil is:
- 1) 37.33
 - 2) 45.33
 - 3) 35.33
 - 4) 43.33
20. The Lassaigne's extract is boiled with conc. HNO_3 while testing for halogens. By doing so it:
- 1) decomposes Na_2S and NaCN , formed.
 - 2) helps in the precipitation of AgCl .
 - 3) increase the solubility product of AgCl .
 - 4) increases the concentration of NO_3^- ions.
21. Artificial sweetener which is stable under cold conditions only is:
- 1) saccharine
 - 2) sucralose
 - 3) aspartame
 - 4) alitame
22. Chloroamphenicol is an
- 1) antifertility drug
 - 2) antibiotic - narrow spectrum
 - 3) antiseptic and disinfectant
 - 4) antibiotic-broad spectrum
23. Which of the following forms cationic micelles above certain concentration?
- 1) Sodium dodecyl sulphate
 - 2) Sodium acetate
 - 3) Urea
 - 4) Cetyltrimethylammonium bromide
24. Antiseptics and disinfectants either kill or prevent growth of microorganisms. Identify which of the following statements is not true?
- 1) Dilute solutions of boric acid and hydrogen peroxide are strong antiseptics.
 - 2) Disinfectants harm the living tissues
 - 3) A 0.2% solution of phenol is an antiseptic while 1% solution acts as a disinfectant
 - 4) Chlorine is used as strong disinfectants

- 1) 1 2) 2 3) 3 4) 4
32. Suppose $A_1, A_2, \dots, A_{30}$ are thirty sets each having 5 elements and $B_1, B_2, \dots, B_n$ are n sets each with 3 elements, let $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^n B_j = S$ and each element of S belongs to exactly 10 of the A_i 's and exactly 9 of the B_j 's. Then n is equal to
- 1) 15 2) 3 3) 45 4) None of these
33. Let A be the set of all determinants of order 3 with entries 0 or 1 only, B is the subset of A consisting of all determinants with value 1, and C is the subset consisting of all determinants with value -1. Then if $n(B)$ and $n(C)$ denote the number of elements in B and C , respectively, we have
- 1) $C = \phi$ 2) $n(B) = n(C)$ 3) $A = B \cup C$ 4) $n(B) = 2n(C)$
34. The minimum number of elements that must be added to the relation $R = \{(1,2), (2,3)\}$ on the set of natural numbers so that it is equivalence is
- 1) 4 2) 7 3) 6 4) 5
35. Let R be a relation on the set N be defined by $\{(x,y) | x, y \in N, 2x + y = 41\}$. Then R is
- 1) Reflexive 2) Symmetric 3) Transitive 4) None of these
36. Consider any set of x_{101} observations $x_1, x_2, x_3, \dots, x_{101}$. It is given that $x_1 < x_2 < x_3 < \dots < x_{100} < x_{101}$. Then the mean deviation of this set of observations about a point x_k is minimum when k is
- 1) 51 2) 50 3) 101 4) 49
37. In an experiment with 10 observations on x , $\sum x = 60, \sum x^2 = 1000$ one observation that was 20 was found to be wrong and was replaced by the correct value 30 then the corrected variance is
- 1) 64 2) 111 3) 52 4) 101
38. In a series of 100 observations, half of them is equal to p and remaining half equal to $-p$ if the standard deviation of these observations is 10 then $|p| = \underline{\hspace{2cm}}$
- 1) 10 2) $\frac{1}{10}$ 3) $\frac{1}{\sqrt{10}}$ 4) $\sqrt{10}$
39. The median and S.D. of a distribution are 25 and 5 respectively if each item is increased by 5 then the new median and S.D. will be
- 1) 20, 10 2) 30, 5 3) 30, 10 4) 25, 10

40. Let $a_1, a_2, \dots, a_n$ be n observations such that $\sum_{i=1}^n a_i = 50$, $\sum_{i=1}^n a_i^2 = 500$ then the possible value of n among the following is _____
- 1) 3 2) 5 3) 4 4) 2
41. The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is
- 1) 40 2) 20 3) 80 4) 60
42. The negation of the statement “ If I become a professor then I will open an IIT ACADEMY
- 1) Neither I will become a professor nor I will open an IIT ACADEMY
 2) I will not become a professor or I will open an IIT ACADEMY
 3) I will become a professor and I will not open an IIT ACADEMY
 4) Either I will not become a professor or I will not open an IIT ACADEMY
43. The converse of “ if two triangles are congruent then they are similar” is
- 1) If two triangles are similar then they are congruent
 2) If two triangles are not congruent then they are not similar
 3) If two triangles are not similar then they are not congruent
 4) None
44. The contra positive of “if in a triangle $ABC, AB > AC$ then $\angle C > \angle B$ ” is
- 1) If in a triangle $ABC, \angle C > \angle B$ then $AB > AC$
 2) If in a triangle $ABC, AB > AC$, then $\angle C > \angle B$
 3) If in a triangle $ABC, \angle C > \angle B$, then $AB > AC$
 4) If in a triangle $ABC, \angle C > \angle B$, then $AB < AC$
45. Consider the following statements.
 P: Suman is brilliant; Q: Suman is rich; R: Suman is honest
 The negation of the statement “Suman is brilliant and dishonest if and only if Suman is rich” can be expressed as:
- 1) $\sim Q \leftrightarrow \sim P \wedge R$ 2) $\sim (P \wedge \sim R) \leftrightarrow Q$ 3) $\sim P \wedge (Q \leftrightarrow \sim R)$ 4) $\sim (Q \leftrightarrow (P \wedge \sim R))$
46. Let p, q, r be statements consider the following
 Statement – I $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ are equivalent
 Statement – II $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \vee (q \rightarrow r)$ are equivalent
 Which of the following is true
- 1) Statement I and II are false
 2) Statement I and II are true

3) Statement I is false and statement II is true

4) Statement I is true and Statement II is false

47. Negation of the statement $p: \frac{1}{2}$ is rational and $\sqrt{3}$ is irrational is

1) $\frac{1}{2}$ is rational or $\sqrt{3}$ is irrational

2) $\frac{1}{2}$ is not rational or $\sqrt{3}$ is not irrational

3) $\frac{1}{2}$ is not rational or $\sqrt{3}$ is irrational

4) $\frac{1}{2}$ is rational and $\sqrt{3}$ is rational

48. If the probability distribution of a random variable x is

$X = x_i :$	-2	-1	0	1	2	3
$P(X = x_i):$	0.1	K	0.2	2K	0.3	K

Then the variance of x is

1) 2.16

2) 2.8

3) $\sqrt{2.16}$

4) $\sqrt{2.8}$

49. If it rains a dealer in rain coats can earn Rs. 500/- a day. If it is fair he will lose Rs 40/- a day. His mean profit if the probability of a fair day is 0.6 is

1) Rs. 230/-

2) Rs. 460/-

3) Rs. 176/-

4) Rs. 88/-

50. A random variable X has the probability distribution

X:	1	2	3	4	5	6	7	8
p(x):	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

For the events $E = \{x: x \text{ is a prime number}\}$ and $F = \{x: x < 4\}$, the probability $P(E \cup F)$ is

1) 0.35

2) 0.77

3) 0.87

4) 0.50

51. For a binomial distribution mean is 6 and S.D. is $\sqrt{2}$. The distribution is

1) $\left(\frac{2}{3} + \frac{1}{3}\right)^9$

2) $\left(\frac{1}{3} + \frac{2}{3}\right)^9$

3) $\left(\frac{1}{5} + \frac{4}{5}\right)^9$

4) $\left(\frac{4}{5} + \frac{1}{5}\right)^9$

52. A student is given 6 questions in a true or false examination. If he gets 4 or more correct answers he passes the examination. The probability that he passes the examination is

1) $\frac{5}{32}$

2) $\frac{7}{32}$

3) $\frac{11}{32}$

4) $\frac{3}{32}$

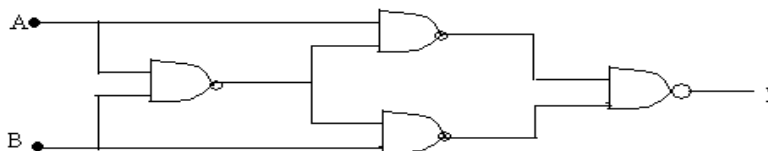
53. 5 cards are drawn one after another successively with replacement from a well shuffled pack of 52 cards. The probability that all the 5 cards are spades
- 1) $\left(\frac{3}{4}\right)^5$ 2) $1 - \left(\frac{3}{4}\right)^5$ 3) $\left(\frac{1}{4}\right)^5$ 4) $1 - \left(\frac{1}{4}\right)^5$
54. Let H be the set of all houses in a city where each house is faced in one of the directions East, West, North, South.
Let $R = \{(x, y) : (x, y) \in H \times H \text{ and } x, y \text{ are faced in same direction}\}$ Then the relation R is
- 1) Not reflexive, symmetric and transitive
2) Reflexive, symmetric, not transitive
3) Symmetric, not reflexive, not transitive
4) An equivalence relation
55. If $10^n + 3 \cdot 4^{n+2} + k$ is divisible by 9, for all $n \in N$, then the least positive integral value of k is
- 1) 5 2) 3 3) 7 4) 1
56. For all $n \in N$, $\{3(5^{2n+1}) + 2^{3n+1}\}$ is divisible by k , where k is prime. Then the least prime greater than k , is
- 1) 13 2) 17 3) 19 4) 23
57. The top of a hill observed from the top and bottom of a building of height 'h' is at angles of elevation 'p' and 'q' respectively. The height of the hill is
- 1) $\frac{h \cot q}{\cot q - \cot p}$ 2) $\frac{h \cot p}{\cot p - \cot q}$ 3) $\frac{h \tan p}{\tan p - \tan q}$ 4) $\frac{h \tan p}{\cot q + \cot p}$
58. An aeroplane flying at a height of 3000m above the ground passes vertically above another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. The height of the lower plane from the ground is
- 1) $1000\sqrt{3}$ m 2) $1000/\sqrt{3}$ m 3) 500 m 4) $1500(\sqrt{3} + 1)$ m
59. A man on the top of a cliff 100 m high, observes the angles of depression of two points on the opposite sides of the cliff as 30° and 60° respectively. Then the distance between the two points is equal to
- 1) $400\sqrt{3}$ m 2) $\frac{400}{\sqrt{3}}$ m 3) $\frac{100}{\sqrt{3}}$ m 4) $\frac{200}{\sqrt{3}}$ m

60. AB is a vertical pole. The end A is on the level ground. C is the middle point of AB. P is a point on the level ground. The portion BC subtends an angle β at P. If $AP = 5AB$ then $\tan\beta =$

- 1) $\frac{5}{51}$ 2) $\frac{5}{24}$ 3) $\frac{5}{26}$ 4) $\frac{5}{28}$

PHYSICS

61. The focal length of a spherical mirror from the following observations: object distance $u = (50.1 \pm 0.5)$ cm and image distance $v = (20.1 \pm 0.2)$ cm is
- 1) (14.3 ± 0.2) cm 2) (14.3 ± 0.1) cm 3) (14.3 ± 0.4) cm 4) (14.3 ± 0.3) cm
62. When a belt moves horizontally at a constant speed of 1.5 ms^{-1} , gravel is falling on it at 5 kgs^{-1} , then, the extra power needed to drive the belt is
- 1) 11.25 W 2) 37.5 W 3) 7.5 W 4) 0.75 W
63. Truth table for system of four NAND gates as shown in figure is:



1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

2)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

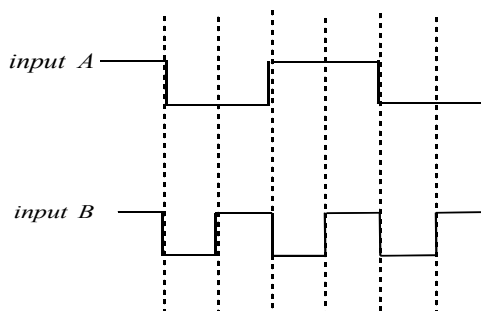
3)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

4)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

64. The input signals at NAND gate is given below



- 1) 2)
- 3) 4)

65. A carrier wave of peak voltage 12 V is used to transmit a message signal. The peak voltage of modulating signal in order to have a modulation index as 0.75 is

- 1) 12V 2) 9V 3) 6 V 4) 3V

66. A message signal of frequency 10kHz and peak voltage of 10 volts is used to modulate a carrier of frequency 1MHz and peak voltage of 20 volts.

a) modulation index= 0.5

b) the side bands produced are at 1010 kHz and 990 kHz

- 1) only a is correct 2) a&b are correct
3) only b is correct 4) both are wrong

67. For a CE transistor amplifier, the audio signal voltage across the collector resistance of $2.0\text{ k}\Omega$ is 2.0 V. Suppose the current amplification factor of the transistor is 100. If the dc base current has to be 10 times the signal current. Then base current is

- 1) 10 mA 2) 1 mA 3) 100 mA 4) 0.1 mA

68. A Plane electromagnetic wave of frequency 25 MHz travels in free space along the x-direction. At a particular point in space and time, $E = 6.3\hat{j}\text{ V/m}$. Strength of the magnetic field at this point

- 1) $-2.1 \times 10^{-8}\hat{k}\text{ T}$ 2) $2.1 \times 10^{-8}\hat{k}\text{ T}$ 3) $2.1 \times 10^{-8}\hat{i}\text{ T}$ 4) $2.1 \times 10^{-8}\hat{j}\text{ T}$

69. A compound microscope is used to enlarge an object kept at a distance 0.03 m from its objective which consists of several convex lenses in contact and has focal length 0.02 m. If a lens of focal length 0.1 m is removed from the objective, find out the distance by which the eye piece of the microscope must be moved to refocus the image.
- 1) 9 cm away from objective 2) 15 cm away from objective
3) 6 cm away from objective 4) 21 cm away from objective
70. An AM wave has 1800 watt of total power content. For 100% modulation, the carrier should have power content equal to
- 1) 1000 watt 2) 1200 watt 3) 1500 watt 4) 1600 watt
71. In a common emitter amplifier, using output resistance of 5000Ω and input resistance of 2000Ω , if the peak value of input signal voltage is 10 mV and $\beta = 50$, then peak value of output voltage is
- 1) $5 \times 10^{-6} \text{ V}$ 2) $12.50 \times 10^{-4} \text{ V}$ 3) 1.25 V 4) 125.0 V
72. If $\vec{a}_1$ and $\vec{a}_2$ are two non-collinear unit vectors and if $|\vec{a}_1 + \vec{a}_2| = \sqrt{3}$, then the value of $(\vec{a}_1 - \vec{a}_2) \cdot (2\vec{a}_1 + \vec{a}_2)$ is
- 1) 2 2) 3/2 3) 1/2 4) 1
73. The temperature of inside and outside of a refrigerator (working based on carnot cycle) are 273 K and 303 K respectively. Assuming that the refrigerator cycle is reversible, for every joule of work done, the heat delivered to the surrounding will be nearly
- 1) 10 J 2) 20 J 3) 30 J 4) 50 J
74. An astronomical telescope has an angular magnification of magnitude 5 for distant objects for normal adjustment. The separation between the objective and the eye-piece is 36 cm and the final image is formed at infinity. The focal length f_o of the objective and f_e of the eye-piece are
- 1) $f_o = 45 \text{ cm}$ and $f_e = -9 \text{ cm}$ 2) $f_o = 50 \text{ cm}$ and $f_e = 10 \text{ cm}$
3) $f_o = 7.2 \text{ cm}$ and $f_e = 5 \text{ cm}$ 4) $f_o = 30 \text{ cm}$ and $f_e = 6 \text{ cm}$
75. If unpolarised light is incident on a crystal at 60° so that the reflected light is completely polarized, then refractive index of the crystal should be
- 1) $\sqrt{2}$ 2) 1 3) $\sqrt{3}$ 4) 1.1

76. A bar magnet of magnetic moment 10 A-m^2 has a cross sectional area of $2.5 \times 10^{-4} \text{ m}^2$. If the intensity of magnetization of the magnet is 10^6 Am^{-1} , the length of the magnet is
- 1) 0.4 m 2) 0.04 cm 3) 0.04 m 4) 40 cm
77. Photo electric emission is observed from a metallic surface for frequencies ν_1 and ν_2 of the incident light ($\nu_2 > \nu_1$). If the maximum value of K.E of the photo electrons emitted in these two cases are in ratio 1: K (K_1 / K_2). Then find the threshold frequency of the metal surface.
- 1) $\frac{K\nu_1 + \nu_2}{K+1}$ 2) $\frac{K\nu_1 - \nu_2}{K+1}$ 3) $\frac{K\nu_2 - \nu_1}{K-1}$ 4) $\frac{K\nu_1 - \nu_2}{K-1}$
78. The input resistance of a transistor is 665Ω . Its base current is changed by $15 \mu\text{A}$ which causes a change of 2 mA in collector current. If this transistor is used as a CE (common emitter configuration) amplifier with a load resistance of $5 \text{ K}\Omega$, what is the voltage gain of the transistor ?
- 1) 1002 2) 1004 3) 1006 4) 1008
79. The electromagnetic waves of frequency 2 MHz to 30 MHz are used
- 1) In ground wave propagation 2) In sky wave propagation
3) In microwave propagation 4) In satellite communication
80. Electric and magnetic field of an electromagnetic wave are
- 1) in phase and perpendicular to each other
2) in phase and parallel to each other
3) in opposite phase and parallel to each other
4) in opposite phase and perpendicular to each other
81. If the amplitude of velocity of a particle acted on by a force $F = F_0 \cos \omega t$ along x -axis is given by $v_0 = \frac{1}{(a\omega^2 - b\omega + c)^{1/2}}$, The resonant frequency is
- 1) $\frac{b}{a}$ 2) $\frac{2b}{a}$ 3) $\frac{b}{2a}$ 4) $\frac{a}{2b}$

82. Two capillary tubes A and B of the same radius r but of lengths $2l$ and l respectively are held horizontally. When a pressure difference P is maintained across A, the rate of flow of water through it is Q . If the tubes are connected in series and the same pressure difference P is maintained across the combination, the rate of flow of water through the combination will be
- 1) Q 2) $\frac{2Q}{3}$ 3) $\frac{Q}{3}$ 4) $\frac{3Q}{4}$
83. A carnot engine takes 3000 kcal of heat from a source at temperature 627°C and gives it to a sink at 27°C . The heat rejected to the sink is
- 1) 2000 k cal 2) 500 k cal 3) 1500 k cal 4) 1000 k cal
84. A telescope has an objective of focal length 120 cm and an eyepiece of focal length 5.0 cm. The magnifying power of the telescope when it is used to view distant objects when the telescope is in normal adjustment is
- 1) 24 2) 12 3) 48 4) 6
85. If I_0 is the intensity of the principle maximum in the single slit diffraction pattern then, with doubling the slit width, the intensity becomes
- 1) I_0 2) $\frac{I_0}{2}$ 3) $\frac{I_0}{4}$ 4) $4I_0$
86. Assuming human pupil to have a radius of 0.25 cm and a comfortable viewing distance of 25 cm, the minimum separation between two objects that human eye can resolve at 500 nm wavelength is
- 1) $1\mu\text{m}$ 2) $30\mu\text{m}$ 3) $100\mu\text{m}$ 4) $300\mu\text{m}$
87. A Polaroid examines two adjacent plane polarized beams A and B whose planes of polarization are mutually perpendicular. In the first position of the analyser, beam B shows zero intensity. From this position a rotation 30° shows that the two beams have same intensity. The ratio of intensity of the two beams I_A & I_B
- 1) 1:3 2) 3:1 3) $\sqrt{3}:1$ 4) $1:\sqrt{3}$
88. A compass needle whose magnetic moment is $60 \text{ A}\cdot\text{m}^2$ pointing geographical north at a certain place, where the horizontal component of earth's magnetic field is $40\mu\text{Wb}/\text{m}^2$, experience a torque $1.2\times 10^{-3} \text{ N}\cdot\text{m}$. What is the declination of the place?
- 1) 60° 2) 45° 3) 30° 4) 90°

89. The N-P-N transistors are preferred over the P-N-P transistors, because
- 1) N-P-N transistors have low heat dissipation
 - 2) N-P-N transistors are cheap and easily available
 - 3) N-P-N transistors have higher mobility of electrons than holes in P-N-P transistors
 - 4) N-P-N transistors can handle large power
90. A light emitting diode (LED) has a voltage drop of 2V across it and passes a current of 10 mA. When it operates with a 6V battery through a limiting resistor R, the value of R is:
- 1) $40\text{ K}\Omega$ 2) $0.4\text{ K}\Omega$ 3) $200\text{ K}\Omega$ 4) $400\text{ K}\Omega$

KEY SHEET

1	2	2	4	3	3	4	1	5	1
6	4	7	2	8	4	9	1	10	2
11	1	12	2	13	4	14	3	15	3
16	4	17	3	18	1	19	1	20	1
21	3	22	4	23	4	24	1	25	3
26	2	27	3	28	3	29	2	30	1
31	4	32	3	33	2	34	2	35	4
36	1	37	4	38	1	39	2	40	2
41	3	42	3	43	1	44	3	45	4
46	3	47	2	48	1	49	3	50	2
51	2	52	3	53	3	54	4	55	1
56	3	57	2	58	1	59	2	60	1
61	2	62	1	63	1	64	2	65	2
66	2	67	4	68	2	69	1	70	2
71	3	72	3	73	1	74	4	75	3
76	3	77	4	78	1	79	2	80	1
81	3	82	2	83	4	84	1	85	1
86	2	87	1	88	3	89	4	90	4